

Pipe radius estimation using Kinect range cameras

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Highlights

- Low-cost range cameras are investigated for 3D geometric modeling.
- A statistical approach is performed for radius detection and estimation from 3D point clouds.
- Optimal nearest neighborhood size for various devices are analyzed.
- Range cameras are capable of detecting and estimating cylindrical objects at close ranges.

Abstract

In the heavy construction industry, pipe spool assembly is geometrically complex and suboptimal fabrication processes inevitably lead to fabrication errors and costly rework. In an attempt to mitigate fabrication risks and improve product quality, computer aided tools are being developed to provide an additional layer of control. In this paper, the data acquisition capabilities of two low-cost range cameras are investigated for the eventual purpose of pipe fitting process monitoring in a smart fabrication facility environment. Range images of various pipes are systematically taken at varying distances from the sensor. By using the proposed radius estimation algorithm, the utility of the data for accurate geometrical pipe feature detection is evaluated by a radius feature metric. Results show that the algorithm reliably extracts radius information from point clouds representing piping. In conjunction with the algorithm, the low-cost range camera hardware was able to characterize pipes, of radius ranging from 2.41 cm to 8.78 cm at a distance from the sensor ranging from 0.5 m to 3.75 m, with an average error of 18% for Kinect 1 and 10% for Kinect 2.

Keywords: pipe spool, radius, Kinect, curvature, fabrication control, assembly control, visual sensing.

1. Introduction

Dimensional compliance assessment of prefabricated and modular assemblies is a critical part of mitigating costly repair and rework on heavy industrial construction projects. As assemblies and modules become more complex, manual direct contact metrology becomes increasingly ineffective at detecting fabrication errors and so automated 3D computer vision alternatives that offer comprehensive data collection must be explored (Dimitrov and Golparvar-Fard 2015; Fathi et al. 2015; Golparvar-Fard et al. 2011; Kim et al. 2016; Nahangi and Haas 2014; Son et al. 2015).

Modular construction is a method of construction that uses three-dimensional blocks or modules that are prefabricated off-site in factory conditions and then transported to and assembled on-site to create complete buildings or industrial facilities. The use of modularization and prefabrication is becoming more prevalent in industrial construction (Bernstein et al. 2011) and as a result, the effective execution of prefabrication will play an increasingly central role in total cost and schedule management on construction projects. Modular pipe spool assembly involves the precise alignment of hundreds of bolts and welds and the fit up of multiple large and dimensionally complex elements; although skilled labor is usually involved, suboptimal fabrication processes inevitably lead to fabrication errors, misaligned module interfaces, and costly rework (Rausch et al. 2017a; Rausch et al. 2017b).

In the construction literature, rework is the wasteful effort involved in redoing work that has not yet yielded a product adequately conforming to contractual requirements (Hwang et al. 2009; Love and Li 2000). Rework directly contributes to cost and schedule overruns on construction projects (Hegazy et al. 2011; Love 2002) and constitutes between 2% and 20% of a typical project's contract amount (CII 2011). Many rework instances during oil and gas turnarounds (TAR's) are attributable to the deficiencies of traditional direct contact metrology and subsequent manual information flow and usage. Ideally, these systems would collect accurate dimensional data of field conditions, reliably present the information to fabricators, and assess whether new and replacement components are fabricated within tolerance of the specified field conditions. However, field and shop measurements taken using measuring tapes, calipers, custom gauges, squares, and straightedges are dependent on accessibility and are prone to measurement errors. Data recorded manually and processed through field sketches, clean drawings, cut length drawings, and finally to new piping is inevitably contaminated with transcription errors. Errors in measurement and transcription increase in severity as the assembly's geometrical complexity increases. Therefore, these traditional processes rely heavily on conscientious craftsmen and cannot consistently facilitate fabrication that mitigates field rework. There is a need for 3D computer vision based tools that offer objective, accurate, fast, and continuous data collection for reliable dimensional compliance control on heavy industrial construction projects. Iterations of these tools have already managed to achieve schedule reductions of 40% and substantial cost savings (Sanders 2001).

2. Background

The related background is investigated from three perspectives: (1) importance of piping in the industrial construction sector, (2) 3D imaging using low-cost range cameras, and (3) pipe radius estimation using vision-based sensors.

2.1. Importance of piping in the industrial construction sector

Pipe spool fabrication falls under the category of industrial construction processes, and involves the creation of piping networks intended to carry water, steam, fluids, gases, chemicals, and fuel as part of heating, cooling, lubricating and process piping systems. On a given project, pipe spool fabrication typically demands the largest percentage of direct labor and is commonly viewed as one of the more complex operations in project execution (CII-Research Team 327).

Fabrication refers to the series of tasks of cutting, bending, forming, fitting, and welding of individual pipe components to each other and their subsequent treatment and examination to form a unit (pipe spool or piping assembly). This process may take place in a commercial pipe fabrication shop or an on-site fabrication shop, where subsets of the piping system are fabricated into subassemblies or modules. When built in commercial pipe fabrication shops away from their final installation site, this is known as off-site fabrication or prefabrication. These off-site fabrication shops typically offer a greater degree of control over fabrication quality and productivity than on-site fabrication. Pipe diameters processed by these facilities range from between 2 inches to in excess of 48 inches. Commercial pipe fabrication shops often perform piping, as well as other architectural and structural metalwork all within the same facility.

During fabrication, all materials and components are assumed to have specific dimensions and the locations of elements are dimensioned on the isometric drawings to a theoretically exact position relative to one or more datum points. In reality, these fabricated dimensions and locations vary somewhat. The contractually acceptable amount of variance is the tolerance for that specific measure. Tolerance specifications assure installation of a system within a reasonable degree of accuracy.

For specifying tolerances, designers and [quality control](#) (QC) specialists typically utilize standardized dimensional tolerance guidelines, such as the systems published by American Society of Mechanical Engineers (ASME) or the International Organization for Standardization (ISO). For any contract, the designer must clearly state: (1) the tolerances allowed, (2) the standards used, (3) how compliance will be verified, and (4) what the result of noncompliance will be.

Currently, the predominant processes for monitoring the critical dimensions outlined in these standards involve manual assessment by certified QC personnel using direct contact measurement devices such as metal measuring tapes, calipers, custom gauges, squares, and straightedges. No measurement process is exact, and this uncertainty needs to be taken into account when verifying compliance with any allowable tolerance. Measurement uncertainty is simply a quantified doubt about the result of a measurement. Uncertainties and errors in measurement can come from many sources, such as the measuring device, the component being measured, the skill of the craft worker measuring, the measurement process, and the environment.

2.2. 3D imaging using low-cost range cameras

The most significant factor in a fabricator's decision to adopt or not adopt a new technology is cost. Therefore, for early stage iterations and adoption, low cost 3D computer vision sensors must be explored for development of improved quality control tools. Microsoft's "range sensing" Kinect sensor was released in November 2010 and is one of the cheapest sensors available in its product class. Kinect was designed as a human-computer game interface, but the sensor's ability to capture 3D data has attracted the attention of researchers in 3D modeling and reconstruction. The Kinect sensor consists of an infrared (IR) projector, an IR camera, and a color camera. The IR projector projects a known infrared-light speckle

pattern into the 3D scene, which is then viewed by the IR camera. The speckle pattern is incident on surfaces of varying depth which make up the scene and is thus distorted. The distorted pattern is compared to the non-distorted reference speckle pattern and a range image is inferred (Han et al. 2013). The resulting point cloud contains about 300,000 points in every frame. The experiment in this paper will make use of these types of low-cost range sensing systems.

The depth image produced by the Kinect sensor contains a constant 640×480 pixels. Considering the point density of the resulting point cloud to be the number of points per unit area, the point density on the XY plane is inversely proportional to the squared distance from the sensor (Khoshelham and Elberink 2012). Therefore, an object being detected by the sensor will be represented by fewer and fewer points as it moves away from the sensor. Results from a Kinect 1 sensor accuracy study concluded that: (1) Random error of depth measurements increases quadratically with increasing distance from the sensor and reaches 4 cm at the maximum sensing distance of 5 m. (2) The depth resolution decreases quadratically with increasing distance from the sensor. The point spacing in the depth direction (along the optical axis of the sensor) is as large as 7 cm at the maximum range of 5 m as can be seen in (Khoshelham and Elberink 2012). Because Kinect 1 has the required accuracy for body skeleton recognition, it has also been used for posture and gesture analysis of construction laborers performing specific tasks (Khosrowpour et al. 2014; Ray and Teizer 2012).

2.3. Pipe radius estimation using vision-based sensors

The development of a 3D computer vision dimensional compliance assessment tool for prefabricated construction assemblies should begin with a minimum viable application (Golparvar-Fard et al. 2011; Kim et al. 2016; Nahangi and Haas 2016; Son et al. 2014; Son et al. 2015). This paper explores pipe radius identification as the minimum viable application of the proposed computer vision system. Once a point cloud has been captured using the sensing device, an algorithm must extract a pipe radius estimate from the point cloud acquired.

The directional curvatures at any point p on a smooth surface M are defined in terms of the curvatures of smooth curves on M containing p . The minimum and maximum directional curvatures computed from the curves are called principal curvatures with corresponding orthogonal principle directions (Manfredo 1976). The first principal curvature k_1 , the principal curvature representing the maximum directional curvature, of any point p on the exterior surface of a pipe corresponds with the pipe's circumference. The radius of the pipe is therefore the inverse of k_1 . [Detailed explanation is provided in Section 3.2.2.](#)

Due to the range sensor's limited depth resolution and resulting noise in the captured data, a probabilistic approach to estimating the pipe's radius is adopted. A probability distribution can be derived, if the principal curvature is calculated for subsets of the larger point cloud surface and then agglomerated. This would require a local surface feature description method that is capable of calculating principal curvatures (directionally discriminant). Yang et al. (2006) assumed that the eigenvectors of the covariance matrices of local neighborhoods serve as principal directions of the given surface at the chosen scale. An et al. (2011) developed a discrete curve model for geometric properties estimation for local discrete surfaces. The algorithm used in this paper is based on the local principal curvature estimation method described in Seibert et al. (2010) and implemented as a local surface feature descriptor for point cloud segmentation in Dimitrov and Golparvar-Fard (2015) as it exhibited the greatest level of robustness.

The experimental setup for testing the low-cost sensor's data acquisition capabilities is similar to Shen et al. (2013).

2.4. Knowledge gap and contribution of the state-of-the-art

The literature investigation showed that pipe spool fabrication plays a critical role in the industrial construction sector. Although accurate and inexpensive sensors have become available recently, inspection and compliance control is usually performed manually, which is tedious, time-consuming, prone to error, and therefore not sufficiently reliable for accurate measurement processes. Integrating reliable and efficient measurement frameworks will substantially improve the efficiency of the process control systems. 3D image-based measurement of pipe spools is relatively limited to the point clouds acquired using terrestrial laser scanning. Although laser scanning is found to be very accurate (Bhatla et al. 2012; Fathi and Brilakis 2013; Golparvar-Fard et al. 2011), it is often not sufficiently effective and efficient because of its cost and the time required for data collection and data processing. This paper presents a preliminary study of a low-cost range camera technology (Microsoft Kinect 1 and 2) and investigates its reliability and measurement capability under various circumstances. The measurement process is limited to pipe radius detection and quantification at this stage. Pipe radius estimation is based on the curvature feature of pipe spools, and the data collection and processing phases are significantly more time-effective compared to the previous approaches. The methodology for this framework is detailed in the following section.

3. Methodology

3.1. Overview of the proposed framework

An overview of the proposed framework for pipe radius estimation capability of Kinect camera is illustrated in Figure 1. As mentioned earlier, the low-cost range cameras investigated in this study are the two Kinect cameras, originally developed by Microsoft for gaming purposes. The radius estimation framework is performed on the 3D images acquired at various distances from the sensors. At each distance, the 3D image is analyzed for investigating the best radius that can represent the cylindrical object sensed in the scene. Once the radius estimations for all of the distances are performed, results are analyzed in order to measure the impact of distance on the accuracy of the radius estimation framework. The pipe radius estimation framework and the required metrics are explained in the following section.

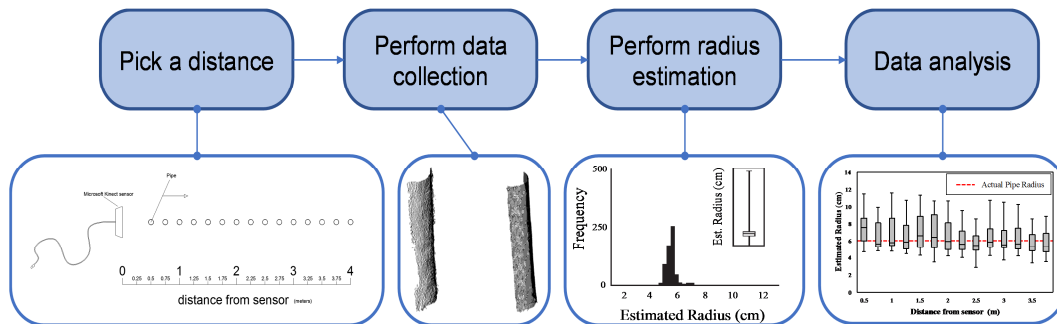


Figure 1: Overview of the proposed framework for pipe radius estimation capability analysis of Kinect 1, Kinect 2, and other low-cost range sensors.

3.2. Radius estimation framework

The radius estimation framework used in this paper is similar to the method presented by the same group of authors (Czerniawski et al. 2016). A summary of the framework is presented in Figure 2. The radius estimation framework has two major components: (1) normal vector extraction, (2) local curvature calculation. These components are briefly explained in the following sections.

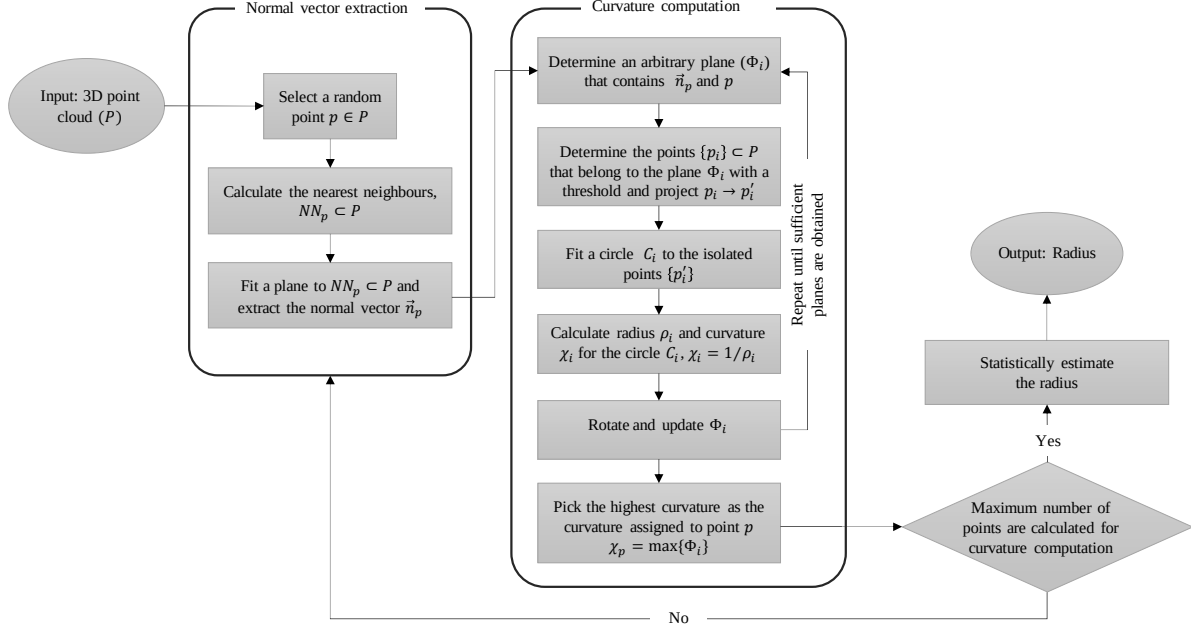


Figure 2: Radius estimation framework has two major modules. Module 1: normal vector estimation (Section 3.2.1), Module 2: curvature computation (Section 3.2.2).

3.2.1. Module 1: Normal vector extraction

The normal vector extraction module is performed in a series of three steps for randomly selected points belonging to an input point cloud, as follows:

- 1- A neighborhood of the randomly selected point is assigned to the investigated point. For calculating nearest neighbors, *kd-tree* space-partitioning is used to expedite the search process.
- 2- A plane is then fitted to the neighborhood containing the originally selected point. Plane fitting is performed using principal component analysis (PCA).
- 3- The normal vector of the fitted plane is stored as the normal vector of the investigated point.

Figure 3 shows a summary of the above-explained three steps for normal vector extraction given an input point cloud (3D image). Once the normal vector is calculated, it is used for calculating the curvature in the following step.

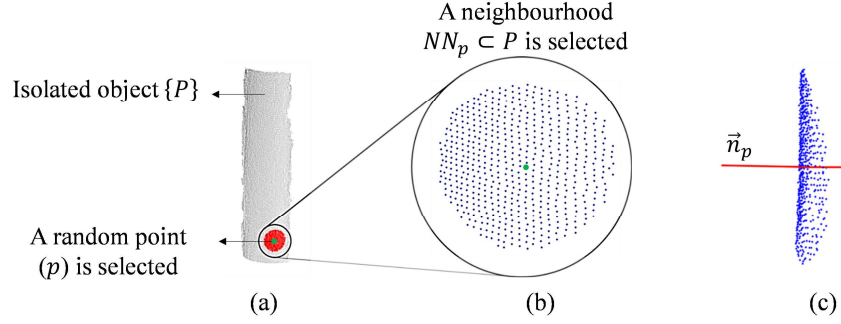


Figure 3: Normal vector extraction module. (a) A random point is selected given an input point cloud; (b) a neighborhood of the randomly selected point is identified; (c) a plane is fitted to the neighborhood containing the originally selected point. See (Czerniawski et al. 2016) for more detailed information.

3.2.2. Module 2: Local curvature/radius computation

Once normal vector ($\vec{n}_p$) at point $p \in P$ is calculated, local curvature (χ_p) at point p is to be computed. For this purpose, a series of five steps is performed as follows:

- 1- An arbitrary plane (Φ_i) containing the normal vector $\vec{n}_p$ and point p is selected.
- 2- The points $p_i \subseteq P$ that sit on or are sufficiently close to the plane Φ_i are computed.
- 3- A circle is fitted to the isolated points $p_i \subseteq P$ and the radius is calculated.
- 4- The plane Φ_i is incrementally rotated and steps 1 to 3 are repeated.
- 5- Minimum radius (maximum curvature) is assigned as the computed local radius (curvature) of the investigated point $p \in P$.

Figure 4 shows a typical example of radius/ curvature at a point $p \in P$. As seen in Figure 2, the procedure is performed with a sufficient number of points in order to statistically estimate a reliable and accurate radius. Eight planes are used in this study similar to previous works (Czerniawski et al. 2016; Dimitrov and Golparvar-Fard 2015). Results are then analyzed and compared to the ground truth measurements using terrestrial laser scanning.

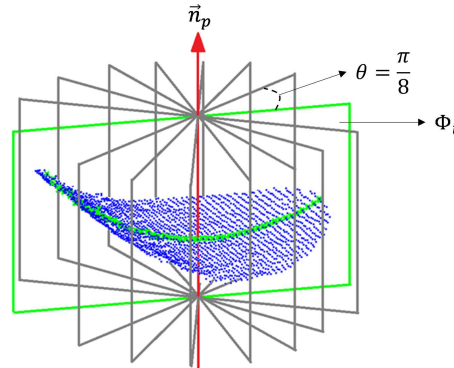


Figure 4: Local curvature/radius computation. An arbitrary plane Φ_i is selected. A circle is then fitted to the points on the point cloud intersected by Φ_i .

4. Experiments and Results

Data acquisition is performed using *Kinect 1* and *Kinect 2*, and a *FARO LS 840 HE* laser scanner. The laser scanner is used to provide contrast and a performance comparison with scanning technology that is two magnitudes more expensive. The true diameters of the pipes are specified by the vendor; however, they were verified using a high precision Vernier caliper. Technical properties and features of the Kinect range cameras used in this study are summarized in Table 1. Range and accuracy specifications for the FARO laser scanner used in this study are summarized in Table 2.

Table 1: Technical properties and features of the Kinect range cameras

Technical Specification	Kinect 1	Kinect 2
core depth sensor technology	infrared (IR) structured light for triangulation	IR indirect time-of-flight illuminator
depth resolution	320×240 ($\approx 75,00$ points) 30 fps	512×424 ($\approx 200,00$ points) 30 fps
field of view	57° horizontal $\times$ 43° vertical	70° horizontal $\times$ 60° vertical
measurement range	40 cm – 6 m	50 cm – 4 m

Table 2: Technical specifications of laser scanner used for data collection (FARO 2014)

Measurement	Range	Accuracy
Distance	0.60-40 m	0.6 mm (at maximum resolution)
Field of View	Horizontal: 360° Vertical: 320°	Horizontal: 0.00900° Vertical: 0.00076°

In order to measure the performance of the proposed framework, four cylindrical pipes are used for the calibration of the curvature computation framework (see Table 3). The first 3D imaging device was positioned at one end of a hallway. As seen in Figure 5, single perspective images were taken of the four pipes starting at a distance from the imaging device of 0.5 m and at each 0.25 m interval to a maximum distance of 3.75 m . The process was repeated for each of the other two 3D imaging devices. A total of 56 images were generated.

Table 3: Properties of the pipes used for the experiments

Descriptor	Pipe 1	Pipe 2	Pipe 3	Pipe 4
ID used for results analysis	P1	P2	P3	P4
Actual radius (cm)	2.41	4.43	6.09	8.78
Length (cm)	40	40	40	40

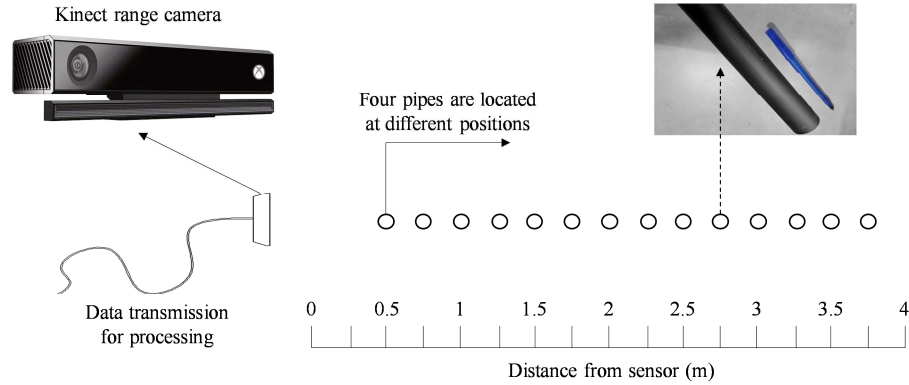


Figure 5: Layout of the experimental setup. Four pipes (Table 3) are located at different distances from the 3D imaging sensor for characterizing the impact of distance on the accuracy of radius estimation.

As discussed, two parameters must be investigated to reliably assess the capability of Kinect cameras for radius estimation: (1) nearest neighborhood size, and (2) radius estimation accuracy. The parameters are investigated, and related typical results and analyses are provided in the following sections.

4.1. Optimal nearest neighborhood size

When running the radius estimation algorithm, the user must know what nearest neighborhood size to specify. The known user inputs are the type of imaging device being used and the curvatures of the search query. In order to explore the relationship between nearest neighborhood size and the accuracy of the associated output, the images collected are manually segmented, and each individual pipe point cloud is analyzed using a series of nearest neighborhood sizes. The accuracy of the curvature characterization is then studied.

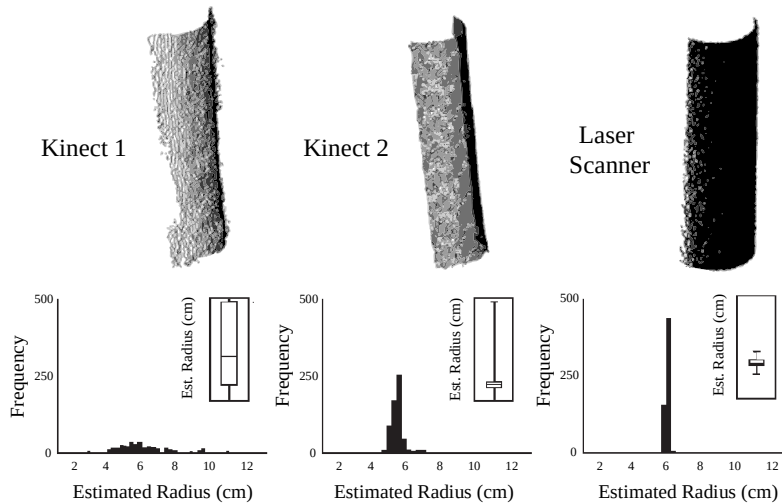


Figure 6: Individual pipe point clouds for P3 at 1.5 m from sensor and generated curvature histograms and distribution box plot

Each of the four pipes was extracted by manually segmenting the collected images resulting in 168 individual pipe point clouds (four pipes at 14 locations and scanned with three devices). The curvature characterization algorithm was run on each of the extracted point clouds using 10 nearest neighborhood sizes (1-10 cm) generating 1680 curvature histograms. The result is a histogram of k_1 curvatures. Figure 6 shows sample curvature histograms for P3 at 1.5 m from the sensors.

The curvature histograms for the entire range of distances are combined into summary charts for each device, pipe radius and nearest neighborhood size combination. Figure 7 shows the 5 cm-radius nearest neighborhood size curvature characterization results for P3 data collected by the Kinect 2 at 0.5-3.75 m from the sensor.

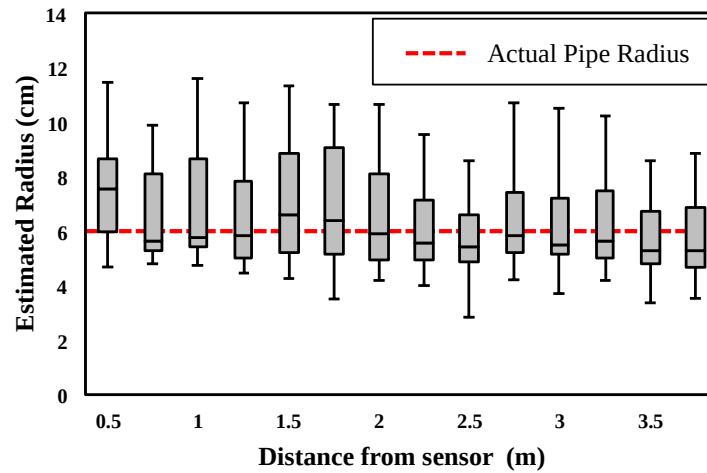


Figure 7: Curvature characterization for P3 with nearest neighborhood size 5 cm (Kinect 2 data source).

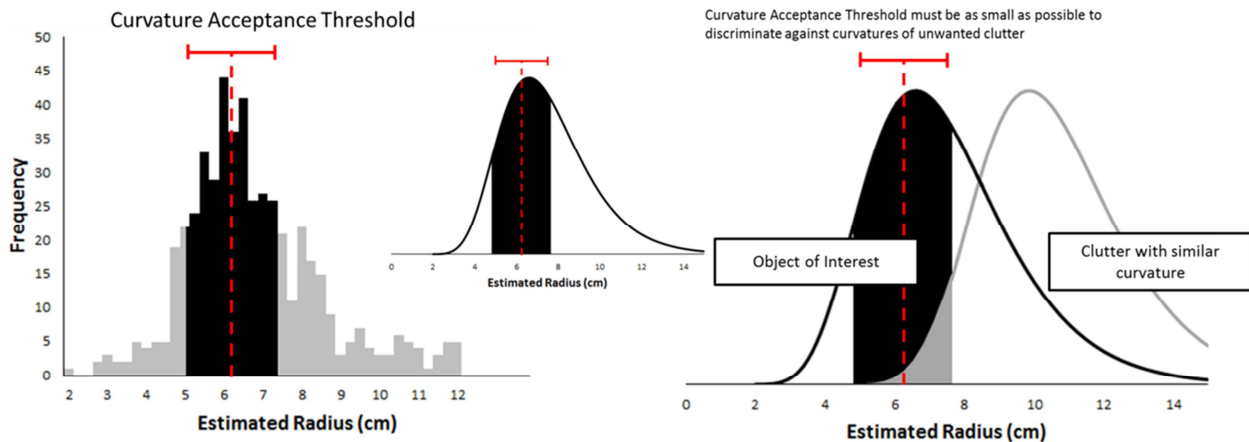


Figure 8: Curvature acceptance threshold specification for Kinect 1 curvature histogram for P3 at 1.5 m.

Understanding that each individual pipe point cloud represents a single curvature type that would be entered as part of a search query in further steps such as the object extraction algorithm (Czerniawski et al. 2016), the histogram must be abstracted to an actual pipe radius (3D model curvature) and curvature acceptance threshold (Figure 8). The curvature acceptance threshold must be as small as possible to discriminate against clutter in the scene that has curvatures similar to those in the search query. For each of the 1680 curvature histograms, the actual pipe radius and curvature acceptance threshold specifications resulting in a 50%-point acceptance rate are calculated. Such acceptance rate implies the median of the distribution. A typical curvature acceptance threshold summary chart is presented in Figure 9. Each boxplot represents a 3D imaging device, pipe radius, and the nearest neighborhood size combination and is comprised of 14 curvature acceptance threshold values (one for each distance 0.5-3.75 m). Using the curvature acceptance thresholds, the optimal nearest neighborhood size is determined for each 3D imaging device and pipe radius combination (Table 4).

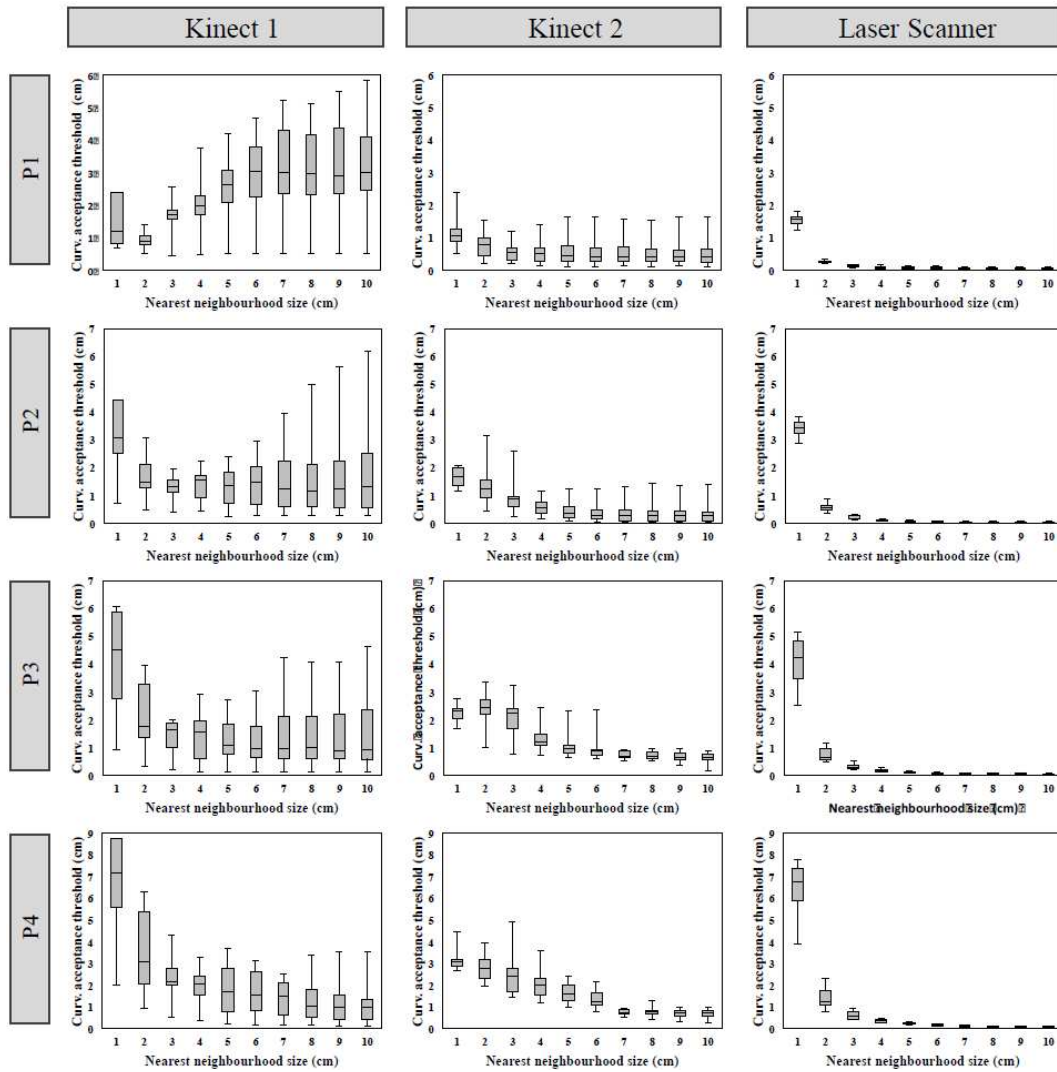


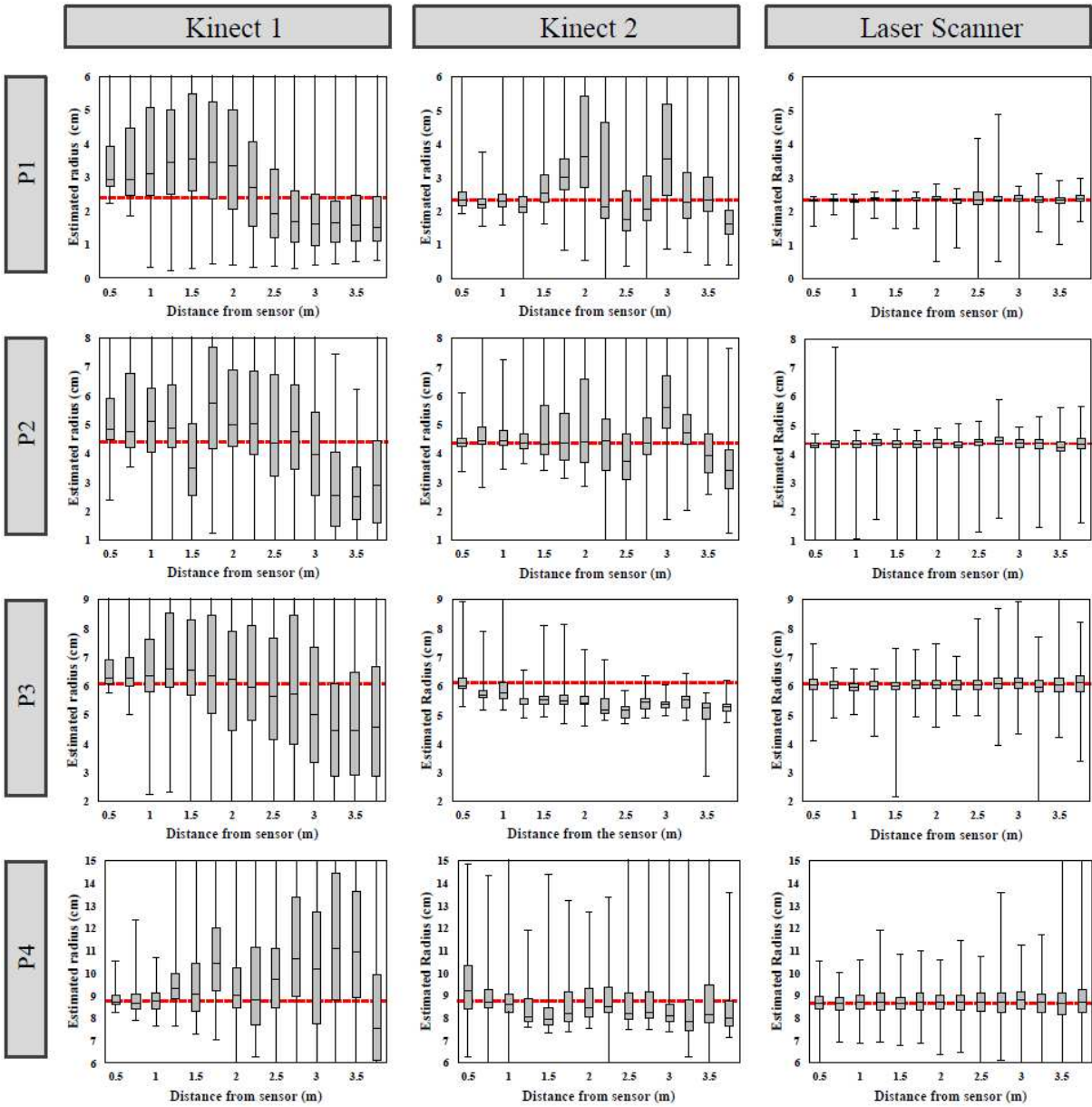
Figure 9: Curvature acceptance threshold for 50%-point acceptance rate summary charts for nearest neighborhood size 1-10 cm, for Kinect 1, Kinect 2, and laser scanner and the pipes.

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Table 4: Optimal nearest neighborhood size for extracting the pipes

3D imaging device	P1	P2	P3	P4
Kinect 1	2 cm	3 cm	3 cm	7 cm
Kinect 2	3 cm	4 cm	10 cm	9 cm
Laser Scanner	4 cm	4 cm	4 cm	4 cm

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Figure 10: Curvature characterization using optimal nearest neighborhood sizes, for Kinect 1, Kinect 2, and laser scanner for the pipes.

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4.2. Radius estimation accuracy

The optimal nearest neighborhood sizes for each 3D imaging device and pipe radius combination were determined by selecting the nearest neighborhood size that yielded the lowest maximum curvature acceptance threshold. The optimal nearest neighborhood sizes for the laser scanner were, however, chosen differently because: (1) the improvement in accuracy after nearest neighborhood size 4 cm was minimal and (2) processing time is quicker for smaller nearest neighborhoods (see Figure 9). The optimal nearest neighborhood sizes are shown in Table 4. The resulting curvature characterizations obtained from implementing the optimal nearest neighborhood sizes are summarized in Figure 10. Using the median as the discrete curvature estimate, the average error of characterization was 18% for Kinect 1, 10% for Kinect 2, and 2% for the laser scanner. The associated curvature acceptance threshold – indicator of curvature estimate spread - was ± 2 cm for Kinect 1, ± 1.1 cm for Kinect 2, and ± 0.25 cm for the laser scanner. Overall, the accuracy of curvature characterization increases as the radius of the pipe of interest increases. As well, the accuracy of the curvature characterization is higher when the pipe of interest is closer to the sensor. The Kinect 1 had a slight bias towards overestimating the radius of the pipes. There was no discernable bias towards overestimation or underestimation by the Kinect 2.

5. Conclusions and Recommendations

Effective dimensional compliance assessment of prefabricated and modular assemblies is a critical part of mitigating costly repair and rework on heavy industrial construction projects. Assemblies and modules are growing in size and complexity, and it is becoming evident that manual direct contact metrology does not have the capabilities to reliably detect all fabrication errors. Thus, automated 3D computer vision alternatives that offer objective, fast, and continuous data collection are being explored. In this paper, a simple computer vision system was developed to detect pipe radius using a Microsoft Kinect sensor and a local point set principal curvature estimation method. The system's ability to accurately estimate pipe radius at different distances from the sensor was studied. The framework used can be applied to evaluate "fit-for-purpose" of any of the dozens of low-cost range sensors that have entered the market in the last few years and that become available in the future.

In order to generate a probability distribution for estimated pipe radius for each point cloud, the point cloud was analyzed locally in different regions, and the estimates from each region were then agglomerated into the histograms. The size of the local regions was one of the algorithm's inputs. Smaller local regions were more susceptible to fitting circles to false local curvatures because of random point cloud undulation due to noise resulting in erroneous radius estimates. Circle fitting on larger local regions had a smoothing effect and was robust to the point clouds noise.

Average error of characterization was 18% for Kinect 1, 10% for Kinect 2, and 2% for the laser scanner. It is predicted that the low-cost systems would be capable of estimating the radius of larger pipes (> 4.4 cm) with greater accuracy. In order to determine the adequacy of these accuracy values, they need to be compared to the specified tolerances within the target control application. Likely, the current iterations of the Kinect cameras will be inadequate for the majority of quality control needs, but are sufficient for coarser progress tracking and positioning applications. *It should also be noted that the Kinect cameras have some limitations for the range and outdoor applications, as they are originally designed for gaming purposes. Their range of scanning is limited to 5 meters and their performance will decrease in outdoor environments because of the other light source interference. The reflectance of the*

pipe is also an important factor for the accuracy of the data collected. A potential avenue of future work may also include the investigation of color and reflectance of the materials used for the pipes.

The presented framework in this paper can be combined with further processing steps such as 3D object extraction and segmentation from cluttered point clouds. As extensively discussed in Dimitrov and Golparvar-Fard (2015) and Czerniawski et al. (2016), the major bottleneck in real-time modeling and processing is the data collection phase. Integrating real-time data collection sensors that are capable of acquiring reliable and accurate data with the construction process control systems will open up an opportunity to substantially improve the processing time. This is currently being investigated at the authors' laboratory at the University of Waterloo. Recommendations for future work include investigating the impact of other critical parameters such as light interference and pipe surface color and reflectance. The radius estimation accuracy should be calibrated under various light conditions and with different colors.

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